Joshua Nee

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Education

Carnegie Mellon University - *Master of Science in Software Engineering* Aug 2025 – Dec 2026
- GPA: 4.0 / 4.0; Research Assistant: Distributed Systems; Teaching Assistant: Computer Systems, Deep Learning

National University of Singapore - *Bachelor of Engineering in Computer Engineering* Aug 2021 – Jan 2025
- GPA: 3.9 / 4.0; Teaching Assistant: Data Structures & Algorithms, Software Engineering & OOP, Machine Learning

Professional Experience

DoorDash May 2026 – Aug 2026
Software Engineering Intern *Sunnyvale, CA*
- Shipped the **first DashPass game backend** a week early for 2,000 weekly players by building Go services, gRPC/Protobuf, Kotlin server-driven UI contracts, Cassandra storage, reward granting, and feature-flag rollout
- Architected the **backend foundation for all future DashPass games**, adopted by 5 games, with reusable execution graphs, Go components, and storage interfaces, reducing per-game registration to 22 lines of code
- Accelerated backend delivery from 6 weeks for the original to 3 days per game by authoring an AI skill that let 2 engineers with no prior context independently build, test, and deploy 2 complete game backends
- Diagnosed 4 silent production defects using distributed traces, uncovering subscription-check inconsistencies during platform migration and resolving win-write failures across iOS and Android without requiring a client release

Shopee Jan 2025 – Jul 2025
Backend Software Engineer *Singapore*
- Implemented fulfillment checks and conditional order creation in Go for **cross-border order synchronization** within a MySQL-backed e-commerce and payment system processing 10M+ daily transactions
- Centralized dependency updates with the infrastructure team by migrating 7 Go services to managed libraries, updating Kafka consumer configurations, and migrating request- and process-level caches
- Restored notification-service deployments in 2 weeks for 100+ internal users as the **service's primary contact**, resolving deployment incompatibilities and repairing its failing test environment
- Engineered backend changes for Singapore's 2025 National Day flag redemption, customizing item/shop redirects and notification email links alongside separate controls to prevent repeat redemptions

ST Engineering May 2024 – Aug 2024
Software Engineering Intern *Singapore*
- Prototyped reusable Apple Vision Pro components in Swift for 3D interactions and live geolocation tracking, **reverse-engineering** undocumented visionOS behavior and reducing future integration time by 30%

InTune Auto Dec 2022 – Dec 2023
Full-Stack Software Engineer *New York, NY*
- Innovated a scheduler for customer self-booking and automated SMS/email confirmations based on shop availability, among 9+ features from a 4-engineer startup later **acquired by Highland Creek Partners**
- Reduced data retrieval latency by 80% on tables with 10K+ rows by redesigning query architecture and implementing keyset pagination in Java, jOOQ, and PostgreSQL for application list views

Projects & Hackathons

A New Agent Hackathon - *LLM Agents* Oct 2025 – Oct 2025
- Won the Vapi Track for a tool-using LLM scheduling agent with text and voice dialogue, integrating ClinicalTrials.gov data, Google Calendar, and Supabase Edge for appointment scheduling

Computer Engineering Capstone Project - *Machine Learning* Aug 2024 – Nov 2024
- Achieved >95% gesture recognition accuracy using a TensorFlow multilayer perceptron and SHAP-guided features, integrating game subsystems via AES-secured TCP/IP and MQTT with packet validation

Technical Skills

Languages & Frameworks: Python, Go, Java, TypeScript, C++, Bash, React, Next.js, gRPC, Protobuf
Databases & Communications: PostgreSQL, MySQL, MongoDB, Firebase, Redis, WebSocket, MQTT, Apache Kafka
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, NGINX, Grafana, Prometheus
AI/LLM: TensorFlow, PyTorch, JAX, LangChain, LangGraph, MCP, RAG pipelines